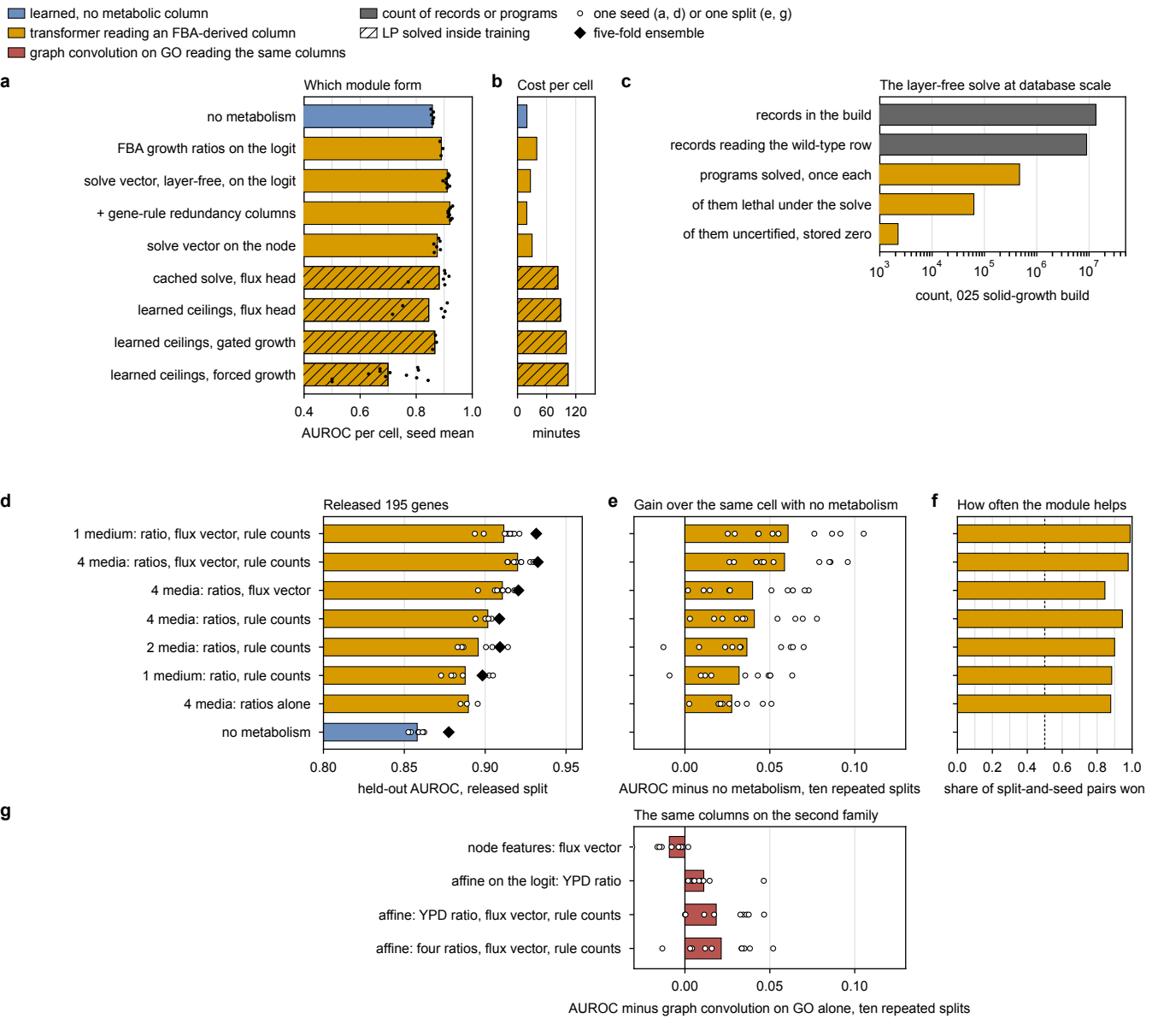

learned, no metabolic column
transformer reading an FBA-derived column
graph convolution on GO reading the same columns
count of records or programs
LP solved inside training
one seed (a, d) or one split (e, g)
five-fold ensemble
a
Which module form
no metabolism
FBA growth ratios on the logit
solve vector, layer-free, on the logit
+ gene-rule redundancy columns
solve vector on the node
cached solve, flux head
learned ceilings, flux head
learned ceilings, gated growth
learned ceilings, forced growth
0.4
0.6
0.8
1.0
AUROC per cell, seed mean
b
Cost per cell
0
60
120
minutes
c
The layer-free solve at database scale
records in the build
records reading the wild-type row
programs solved, once each
of them lethal under the solve
of them uncertified, stored zero
10^3
10^4
10^5
10^6
10^7
count, 025 solid-growth build
d
Released 195 genes
1 medium: ratio, flux vector, rule counts
4 media: ratios, flux vector, rule counts
4 media: ratios, flux vector
4 media: ratios, rule counts
2 media: ratios, rule counts
1 medium: ratio, rule counts
4 media: ratios alone
no metabolism
0.80
0.85
0.90
0.95
held-out AUROC, released split
e
Gain over the same cell with no metabolism
0.00
0.05
0.10
AUROC minus no metabolism, ten repeated splits
f
How often the module helps
0.0
0.2
0.4
0.6
0.8
1.0
share of split-and-seed pairs won
g
The same columns on the second family
node features: flux vector
affine on the logit: YPD ratio
affine: YPD ratio, flux vector, rule counts
affine: four ratios, flux vector, rule counts
0.00
0.05
0.10
AUROC minus graph convolution on GO alone, ten repeated splits